Silence Is Not Violation: Measuring and Reducing False Violation Errors in LLM Compliance Checking of Vendor Data Processing Agreements under India's DPDP Act

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Abstract—Large language models (LLMs) are increasingly used to check commercial contracts against regulations. We identify and measure a recurring failure in which a checker that judges each clause in isolation reports a contract's silence on a requirement as a violation of it. We observed this failure while building PactScan, an LLM- and knowledge-graph-based compliance system for vendor Data Processing Agreements (DPAs) under India's Digital Personal Data Protection (DPDP) Act, 2023, where a fully compliant contract received 19 findings. We release a minimal-pair benchmark of 138 synthetic DPAs (1,752 graded contract–requirement pairs) in which each contract follows, omits or breaks exactly one obligation, including "trap" contracts whose amendments hide or cure a violation. We compare the original clause-level checker with a method that (i) instructs the model that silence is neutral, (ii) checks each requirement only against clauses on its subject, (iii) reads each clause together with the provisions that override it, and (iv) decides at the level of the whole contract. On gpt-oss-120b the false violation rate falls from 60.9% to 19.6% while no mandatory gap is missed and 99% of violations are still caught. The same pattern holds on gpt-oss-20b (from 60.8% to 27.1%) and qwen3.8-27b (from 61.2% to 14.1%), in all 18 company and model units of a conservative sign test, and in each of five independent runs. An ablation shows that the fix comes from the unit of decision rather than from the prompt, which alone barely helps. When existing contracts are re-checked against new regulations, the contract-level rule also misses no gap (F1 76.5%). Residual false alarms concentrate on compound requirements that bundle several duties into one rule.

Index Terms—large language models, legal NLP, contract compliance, natural language inference, DPDP Act, benchmark

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I. Introduction

Companies sign Data Processing Agreements (DPAs) with every vendor that handles personal data on their behalf, and India's Digital Personal Data Protection (DPDP) Act, 2023 [1] makes the company (the Data Fiduciary) accountable for what its vendors (Data Processors) do. Reviewing a portfolio of such contracts against the Act, an internal playbook and, for many firms, the EU General Data Protection Regulation (GDPR) [2] is slow and repetitive, which makes it an obvious target for automation with large language models (LLMs).

A natural design, and the one from which most systems start, is to split the contract into clauses and ask a model, for every clause and every relevant requirement, whether the clause entails, contradicts or is neutral to it, in the style of contract natural language inference (NLI) [3]. This paper studies a failure of that design that is easy to miss and expensive in practice, in which a clause that simply does not deal with a requirement is reported as violating it. For example, a cross-border-transfer clause is flagged for not restating breach-notification duties, and a confidentiality clause for never mentioning children's data. Each alarm can be defended in isolation ("this clause does not secure X"), but it is wrong for the contract as a whole, because another clause secures X.

We first encountered this failure during the development of PactScan, a working compliance platform that parses vendor DPAs into a Neo4j knowledge graph and then checks them with an LLM. Its first version raised 19 findings on a contract that we had written to be fully

compliant. Later, a regulation-update feature produced 9 and then 13 findings for one amendment on a two-page contract, and a first fix that reported only explicit contradictions hid contracts that never mention a newly mandatory protection.

Contributions. (1) We characterise the silence-as-violation error and its opposite failure, the missed gap, and define a pair of metrics that measure both. (2) We release a minimal-pair benchmark of 138 synthetic vendor DPAs under the DPDP Act, with GDPR as a second standard, containing 1,752 graded contract–requirement pairs and trap contracts in which an amendment hides or cures a violation. To our knowledge it is the first compliance benchmark for DPDP contracts. (3) We evaluate a four-part method through a step-by-step ablation on three open models, using conservative statistics. (4) We show that the decisive component is the unit of decision, not the prompt, and that residual errors concentrate on compound requirements.

II. Related Work

Contract understanding and NLI. CUAD [4] provides expert clause annotations for 510 contracts, and we use its category names only as a taxonomy reference. ContractNLI [3] treats contract review as document-level NLI with an explicit NotMentioned label, which plays the role that unanswerable questions play in reading comprehension [5]; our metrics rely on the same three-way distinction. LegalBench [6] collects legal-reasoning tasks. CLAUSE [7] embeds ten categories of flaws in CUAD and ContractNLI contracts and reports that LLMs often miss subtle discrepancies. Our benchmark differs in jurisdiction, document type and purpose, since it separates silence from violation through small, controlled edits whose correct effect is known beforehand, as in CheckList [8] and contrast sets [9].

Compliance checking. Amaral et al. [10] and Azeem and Abualhaija [11] check GDPR DPAs for completeness, and Hassani et al. [12] argue that LLM-based compliance analysis needs more context than a single sentence. They treat completeness (what is missing) and conformity (what is wrong) as separate questions, which a clause-level checker conflates. Work on the DPDP Act checks website privacy policies with access-control rules [13] or embeds compliance logic in software agents [14], and we found no study of DPDP contracts. LLMs hallucinate on legal questions [15], also inside retrieval-based legal research tools [16]. Compliance verdicts are often unchanged when the governing rule is deleted, swapped or negated [17], and knowledge-graph-augmented gap detection has been studied for financial institutions' policies [18]. Our study complements this work, since we keep the rule fixed and vary how verdicts are turned into findings.

III. The System and the Problem

A. PactScan's Detect pipeline

PactScan parses an uploaded contract, extracts clauses with an LLM into a fixed taxonomy (e.g. Breach Notification, Data Processing) and writes them into a Neo4j knowledge graph. When a clause contains override language ("notwithstanding", "supersedes"), an LLM confirms whether it changes another clause, and an Overrides edge is stored. Detect then checks each clause against the requirements relevant to its type, namely 8 internal playbook rules and 5 regulation clauses (DPDP Sections 4, 8 and 9; GDPR Articles 6 and 8). Each check is a single batched structured-output call that returns, for every requirement, an Entails/Contradicts/Neutral label together with a confidence and an explanation. The findings then go to human-approved redlining, which this paper does not touch. The original Detect prompt told the model that "a clause that is vague or silent on a specific mandatory protection … CONTRADICTS the requirement rather than being NEUTRAL", and its decision rule turned every contradiction, and every neutral verdict given with high confidence, into a finding.

B. Three outcomes, two errors

For a contract c and requirement r, the correct outcome is NONE when nothing is wrong (some clause satisfies r, or r does not bind c), VIOLATES when a clause goes against r, and GAP when r is mandatory and nothing in c addresses it. A checker flags (c, r) if it produces at least one finding for r in c. The false violation rate (FVR) is the share of NONE pairs that are flagged, and it is the error that the paper addresses. The missed gap rate (MGR) is the share of GAP pairs that are not flagged, which is the kind of error an over-cautious fix creates. We also report violation recall (share of VIOLATES pairs flagged) and precision, recall and F1 over all problems (VIOLATES $\cup$ GAP).

IV. Method

We compose four changes, each addressing one way silence leaks into findings (Fig. 1).

(1) Conservative prompt. The model is told that a clause that "does not deal with this requirement's subject" or is "merely silent on an adjacent obligation" is Neutral, and that Contradicts is reserved for a clause that itself creates the problem.

(2) Subject routing. Each requirement is checked only against clauses of the type that governs its subject (e.g. breach notification only against Breach Notification clauses), instead of, say, every data-processing clause against every regulation.

(3) Override-aware reading. A clause is classified on its effective text (ET), that is, its own text followed by every provision that overrides it, found by traversing Overrides edges up to three hops in the graph.

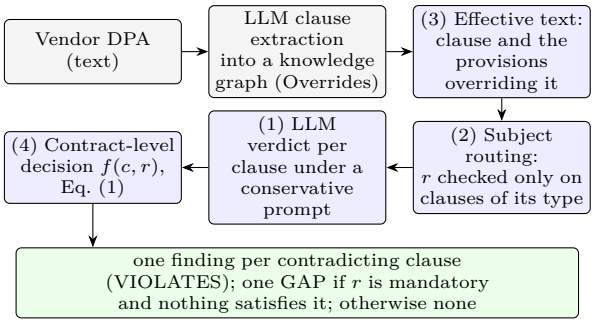


Fig. 1. The checking pipeline for a contract c and each of the 13 requirements r (playbook, DPDP, GDPR). Numbers mark the four changes studied here; configuration A uses none of them and D uses all four. The evaluation feeds the benchmark's gold clauses and override links into step (3), so only the verdict and decision steps are measured.

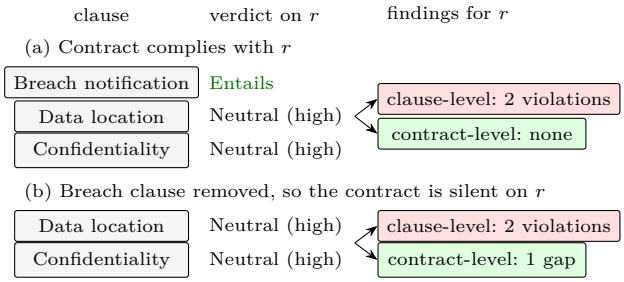


Fig. 2. Two decision rules applied to the same verdicts for a mandatory requirement r ("notify breaches within 72 hours"). In both contracts the clause-level rule reports silence as a violation, whereas the contract-level rule reports nothing in (a) and a single gap in (b).

(4) Contract-level decision. Let $V(c, r)$ be the multiset of verdicts for r over the clauses of c routed to r. The number of findings is

$$f(c,r) = \begin{cases} \#\{\text{Contradicts} \in V\} & \text{if one contradicts } r, \\ 0 & \text{else if one entails } r, \\ 1 & \text{else if } r \text{ mandatory}, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

A violation is still reported on the clause that causes it, while silence becomes at most one gap finding, and only for a mandatory requirement that nothing in the contract satisfies (Fig. 2). Whether a requirement is mandatory is a property of the rule set by the playbook author, not of the contract.

Configurations. We add the changes one at a time. A is the original checker (original prompt, broad routing, clause-level rule); B is A + conservative prompt; C is B + subject routing; D is C + contract-level decision (the full method). D−ET is D without override-aware reading. P is PactScan's Detect agent in its deployed form, run without modification, and its gap logic is limited to the two children's-data requirements. All configurations use the same batched call, client, temperature and model.

V. Benchmark

Construction. Six base DPAs pair a fictional Indian company with a vendor in different sectors (health-tech, retail, ed-tech, fintech, HR/payroll, logistics), differing in parties, data categories, locations, notification windows and liability floors, and written in three drafting styles. Each base contains nine topic clauses (breach notification, liability cap, processing details, children's data, sub-processors, data location, deletion on termination, audit rights, lawful basis) and four filler clauses, and complies with all 13 requirements. Each base yields 23 contracts, namely the base itself, nine silent variants (one topic clause removed), nine breaks variants (one topic clause rewritten so that it breaks its requirements) and four trap variants for breach notification and deletion, in which a later amendment either hides a violation behind a compliant clause or cures a non-compliant one. Every variant differs from its base in exactly one place, so any change in a checker's output is attributable to it. With 13 requirements (8 playbook, 3 DPDP, 2 GDPR), the 138 contracts give 1,752 graded pairs (1,599 NONE, 96 VIOLATES, 57 GAP), and 12 of the contracts hide a violation and 12 hide a fix. The checker is given the benchmark's own clauses, clause types and override links, so parsing and extraction errors are excluded and only the verdict step is measured.

Labels. Each requirement belongs to one topic, so its correct answer follows from which topic was changed and in what way. Seven of the nine topics are mandatory. Silence on the other two breaks nothing, because a contract without a liability cap has unlimited liability (the rule forbids only a cap that is too low) and having a lawful ground is the Data Fiduciary's own statutory duty. The children's-data rules do not bind the payroll base, which processes only employees' data; the playbook requires every other base to address children's data explicitly, so silence on it counts as a gap. Before any model was run, a clause-by-clause review of all labels led us to rewrite three breaking clauses that touched another topic's rule and to exclude 42 pairs on which careful readers could disagree.

VI. Experimental Setup

Models and controls. We use gpt-oss-120b and gpt-oss-20b [19] and qwen3.8-27b [20], served by GroqCloud at temperature 0.2, with each run pinned to a single model and no fallback. Every run has its own response cache, so repeated runs never reuse one another's answers, and within a run each distinct question under a given system prompt is answered once.

Statistics. Confidence intervals (95%) are obtained from a bootstrap [21] over contracts with 2,000 resamples, and per-pair correctness is compared with exact McNemar tests [22]. These figures overstate certainty, since variants of one base share 12 of their 13 clauses and their pairs are therefore not independent. As a more conservative check we also report a sign test that treats each (base,

model) combination as a single unit. Agreement across runs and models is reported as the share of pairs answered identically and Fleiss' κ [23].

Experiments. E1 runs all configurations on gpt-oss-120b over all 138 contracts. E2 runs A and C/D on the two other models. E3 runs A and C/D five times on two bases (46 contracts) on gpt-oss-20b, since E1 and E4 used up the free daily quota for gpt-oss-120b. E4 re-checks contracts against four simulated DPDP amendments.

VII. Results

A. E1: the error and the ablation

Table I shows the error in full. The original checker flags 60.9% of the pairs on which nothing is wrong and raises 21.7 findings on a contract that complies with everything, which is close to the 19 observed during development. It also misses half the gaps, because for a removed clause a mandatory requirement often has no clause of a routed type left to ask about. The full method reduces the false violation rate to 19.6%, misses no gap, keeps violation recall at 99% and raises 2.5 findings on a clean contract. The difference between A and D is significant on matched pairs (691 pairs correct only under D vs. 1 only under A; McNemar $p = 6.7 \times 10^{-206}$) and under the sign test in E2.

Effect of the prompt. Moving from A to B changes the false violation rate by only one point (McNemar $p = 0.004$), although the prompt changes the model's behaviour dramatically. The share of Contradicts/Entails/Neutral verdicts is 65.0/30.9/4.0% under A, 7.4/28.2/64.4% under B and 10.1/40.3/49.6% under C/D. Under the clause-level rule, however, a neutral verdict given with high confidence is still turned into a finding, and so silence that the prompt has correctly relabelled ends up being reported anyway. Subject routing (C) removes some of these checks (51.7%), and the contract-level decision (D) removes the rest, with 549 pairs becoming correct and none becoming incorrect between C and D ($p = 1.1 \times 10^{-165}$).

Effect of override-aware reading on hidden changes. In trap contracts where an amendment hides a violation, D catches 94.4% of the violated pairs, compared with only 27.8% when override-aware reading is removed ($p = 8.0 \times 10^{-7}$ on trap pairs). Where an amendment instead cures a clause, reading the clause alone wrongly flags every such pair (100%), against 33.3% with override-aware reading.

Comparison with the deployed system. P raises the fewest alarms (14.1%) but misses 80.7% of gaps, since its gap logic covers only the children's-data rules. Overall pair accuracy does not differ significantly between D and P ($p = 0.21$), yet their errors run in opposite directions. D accepts a few extra alarms for complete gap coverage, the better trade-off when a missed gap costs more than a false alarm.

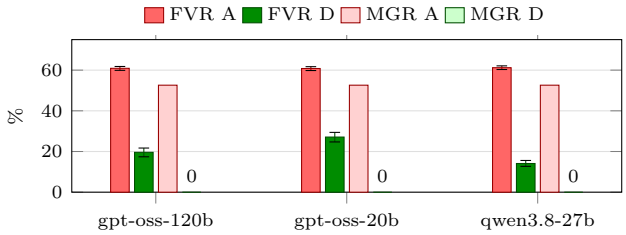


Fig. 3. E2: false violation rate (FVR) and missed gap rate (MGR) of the original checker A and the full method D on three models, over all 138 contracts. Error bars are 95% bootstrap intervals for FVR.

B. E2: Behaviour across models

Under A, all three models make the error at the same rate (60.8–61.2%), which places its source in the design of the checker rather than in any particular model. Under D, every model improves and none of them misses a gap (Fig. 3). On gpt-oss-20b the false violation rate falls from 60.8% to 27.1% with violation recall at 99.0% under D (100.0% under A), and on qwen3.8-27b from 61.2% to 14.1% with violation recall at 88.5% (99.0% under A).

McNemar tests give $p < 10^{-168}$ for each model, and under the conservative test D has a lower false violation rate than A in 18 of 18 (base, model) units (exact sign test, $p = 7.6 \times 10^{-6}$). Qwen has the lowest false violation rate but also catches only 38.9% of hidden violations in trap contracts, so model-specific weaknesses remain.

Under A the three models answer 99.3% of pairs identically (Fleiss' $\kappa = 0.99$) because they all flag nearly everything, whereas under D they agree on 82.0% ($\kappa = 0.70$), so agreement among models is not evidence of correctness.

C. E3: Stability across runs

We ran A and C/D five independent times on gpt-oss-20b over the 46 contracts of two bases (584 graded pairs), each run with its own empty cache. All five runs agree on 99.3% of pairs under A (Fleiss' $\kappa = 0.99$), 83.4% under C ($\kappa = 0.85$) and 78.4% under D ($\kappa = 0.79$), and the per-run false violation rate ranges over 60.5–61.1%, 45.9–56.6% and 22.6–30.8% respectively. In every run D's false violation rate is about half of A's or lower (30.8% against 60.9% in the worst run; mean 26.5% against 60.8%), D misses no gap in any run, and its violation recall stays between 96.9% and 100%. A is stable because it flags nearly everything, while under D 21.6% of pairs change answer in at least one of the five runs. As agreement across samples tracks a model's reliability [24], findings that change on a re-run are best sent to human review.

D. E4: Re-checking contracts when the law changes

PactScan's regulation-update path embeds a new rule, retrieves each contract's related clauses (top-15 by cosine similarity, threshold 0.35) and classifies them against it. We simulated four DPDP amendments, namely a 72-hour processor breach-notice duty, erasure within 30 days

TABLE I
E1 on gpt-oss-120b, 138 contracts (%; 95% bootstrap CI in brackets). "Clean" is the mean number of findings on the six base contracts, which are fully compliant.

Configuration	False violation rate	Missed gap rate	Violation recall	F1	Clean
A original checker	60.9 [59.9, 61.8]	52.6 [38.6, 69.8]	100.0	19.7	21.7
B + conservative prompt	59.9 [58.9, 60.9]	52.6 [38.6, 69.8]	100.0	19.9	21.8
C + subject routing	51.7 [50.4, 52.8]	63.2 [46.4, 81.2]	99.0	21.2	12.5
D + contract-level decision (full)	19.6 [17.4, 21.7]	0.0	99.0	49.2	2.5
D−ET full, no override-aware reading	20.3 [18.1, 22.5]	0.0	86.5	45.3	2.5
P PactScan as deployed	14.1 [12.1, 16.1]	80.7 [67.7, 95.2]	77.1	36.6	1.8

TABLE II
E4: re-checking 46 contracts against four simulated DPDP amendments (gpt-oss-120b, 184 checks, %).

Rule	FVR	MGR	F1	Findings/check
Every non-entailing clause	100.0	0.0	50.4	12.15
Contradictions only	31.1	96.2	21.4	0.31
Contract-level (final)	31.1	0.0	76.5	0.58

TABLE III
False violation rate of D per requirement (%; requirements with 0% on all models omitted).

Requirement	120b	20b	qwen
DPDP s8 (security, accuracy, breach notice)	84.3	100.0	52.8
GDPR Art. 8 (child consent, age threshold)	84.3	65.4	81.9
DPDP s9 / playbook: parental consent	19.7	17.3	17.3
DPDP s4 (grounds for processing)	16.7	65.9	0.0
GDPR Art. 6 (lawful basis)	16.7	31.0	0.0
Playbook: liability cap	15.9	15.9	15.9
Playbook: processing details	1.9	50.9	0.0

with written confirmation, a transfer restriction that also requires naming storage locations, and a new duty to name a grievance contact in India. We then re-checked the 46 contracts of two bases (184 checks: 122 NONE, 10 VIOLATES, 52 GAP) under the three rules the system actually went through (Table II).

The first rule reproduces the development incident, flagging every compliant contract with 12.15 findings per check on average. Restricting findings to contradictions removes 69 points of false alarms but misses 96.2% of gaps, among them every contract that is silent on the new grievance-contact duty. The contract-level rule keeps the lower false alarm rate and recovers every gap (F1 76.5%). Violation recall is 100% under all three rules and retrieval (12.8 related clauses per check) surfaced the relevant clause for every real violation, so the errors come from the verdict and the decision rule.

For the erasure amendment, 36 of 40 compliant contracts were flagged because the model marked the processing-details clause (data deleted within 30 days) as contradicting the amendment for not also requiring written confirmation, although the deletion clause does require certified deletion. Under the contract-level rule this incomplete clause outvotes the one that satisfies the rule, so the silence-as-violation error reappears inside the Contradicts label.

E. Where the remaining false alarms come from

D's remaining false alarms are not spread evenly (Table III). Five playbook rules that each state a single duty (breach-notification window, sub-processor approval, data location, deletion on termination, audit rights) have a false violation rate of 0% on every model. The worst two are compound requirements. DPDP Section 8 bundles security measures, data accuracy and breach notification, and the models mark a compliant breach clause as contradicting it because it does not also address data accuracy, which is a duty of the Data Fiduciary rather than of the processor. GDPR Article 8 includes an age threshold, and the models mark a clause requiring verifiable parental consent for every child as incomplete because it does not state the threshold, although it is stricter than the article. Both are the silence-as-violation error inside a single requirement, and without them D's false violation rate on gpt-oss-120b would be 8.4%. For every model, the hardest base is the ed-tech contract, which is the only one that actually processes children's data (D: 44.4% on gpt-oss-120b, compared with 10.8–16.2% for the other bases).

VIII. Discussion

Choice of decision unit. The error comes less from what the model says than from how its answers are aggregated, so a better prompt changes the model's words but not the output of a clause-level rule. Self-reported confidence does not rescue that rule either, although models can partly estimate their own correctness [25], since the original rule used a high-confidence Neutral as a trigger and so turned correctly relabelled silence back into findings. In PactScan's regulation-update feature, changing what is flagged (every non-entailing clause → contradictions only) simply exchanged one error for another, whereas changing the unit flagged (clause → contract) fixed both.

Next steps. Splitting each requirement into atomic duties and aggregating per duty would stop silence on one duty from contradicting another, and checking each contradiction against the contract's satisfying clauses would address the E4 pattern. For the deployed system, extending the contract-level gap rule to every mandatory

requirement would close its 80.7% missed-gap rate at the cost of a few extra alarms.

IX. Threats to Validity and Limitations

Synthetic contracts and scope. The benchmark is synthetic (six bases, three drafting styles), and real DPAs are longer and messier. Since clause types and override links are given, 36 of the 57 gaps have no clause of their subject left and are decided without the model; all 21 gaps decided from model verdicts were caught. End-to-end accuracy with automatic extraction is not measured.

Labels. Labels were written by the project team, not by practising lawyers, and no second human annotator has been measured yet; we release a blind sheet of 120 stratified pairs over 83 anonymised contracts with a script that reports Cohen's κ [26]. As an automated audit, a model used in no experiment (gemini-3.5-flash-lite, temperature 0) labelled the sheet from the same instructions and agreed on 101 of 120 pairs ($\kappa = 0.73$), including all 60 None pairs. None of the 19 disagreements revealed a label error. Treating two topics as non-mandatory and applying GDPR regardless of territorial scope are stated assumptions.

Settings. We used three open models on one provider at a single temperature, free-tier quotas limited E3 and E4 to two bases, and E4's amendments are simulated.

X. Conclusion

Clause-by-clause LLM compliance checking mistakes silence for violation, and on a matched benchmark of DPDP vendor DPAs every model tested flagged about 61% of the pairs on which nothing is wrong. A conservative prompt changes the model's verdicts but not this outcome. Deciding at the level of the contract, with subject routing and override-aware reading, cuts false violations to 14–27% across models, misses no mandatory gap and keeps violation recall at 88–99%, and the gain repeats across runs and when contracts are re-checked against new regulations. The remaining errors point to compound requirements and incomplete clauses read as contradictions. Future work will test whether approved AI-drafted redlines resolve their findings, whether stated confidence predicts wrong verdicts, and a larger, lawyer-labelled benchmark. The benchmark, harness, raw verdicts and scoring scripts are available at https://github.com/saurabh-1702/PactScan-AI (branch research/silence-not-violation).

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